Lesson notes · for you

The middle and the spread

Data science · Level 1 · Lesson 2

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Work out the mean, median and mode.
- See how one extreme value pulls the mean.
- Measure the spread with the range.

Activity: Find the middle

1. Warm-up (5 min): Line up by birthday month and find the person in the middle of the line.
2. Collect (10 min): In pairs, collect walking times from ten classmates.
3. Work out (10 min): Sort the numbers, find each result and decide which middle fits.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Why did one big number move the mean?
- Which middle did you choose?
- What does the range add?

Check yourself

- For 2, 4, 4 and 10, write the mean, the median, the mode and the range.

Try it this week
Ask ten friends how many minutes they walk to school. Work out the mean, the median and the range, and say which middle describes them best.

Worksheet

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Name _____ Class _____ Date _____

1 **A class has a range of 18 marks. What does that tell you?**

◯ Eighteen pupils passed ◯ The marks are very spread out ◯ The average is 18

2 **Use the numbers 2, 4, 4, 5, 10.**

The median is _____ and the range is _____.

Word bank: 5 · 4 · 8 · 10

3 **Five friends walk 10, 12, 8, 10 and 60 minutes to school. What is the mean?**

◯ 20 minutes ◯ 10 minutes ◯ 12 minutes ◯ 100 minutes

4 **Match each word to how you find it.**

Mean	• • • •	The most common one
Median	• • • •	Add up and divide
Mode	• • • •	The middle one in order

5 **One friend walks 60 minutes; everyone else walks about 10. Which describes most of you best?**

◯ The biggest number ◯ The mean ◯ The median

✎ **When have you heard an "average" that didn't describe most people? Which middle would have been better?**

Check your answers

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1 **The marks are very spread out**

The range is the highest mark minus the lowest: it measures spread, not the middle.

2 **The median is 4 and the range is 8.**

The middle of five ordered numbers is the third, 4.

3 **20 minutes**

Add them up: 10 + 12 + 8 + 10 + 60 = 100, and 100 ÷ 5 = 20 minutes.

4 **Mean — Add up and divide**
Median — The middle one in order
Mode — The most common one

Three different ways to find a typical value.

5 **The median**

The big number pulls the mean up, but the median stays in the middle.